

TOPIC : WORK, ENERGY & POWER

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A.	1	3	4	1	2	2	2	4	2	4	1	1	2	4	2
Q.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	2	3	2	3	1	1	2	3	1	1	2	3	2	4	2
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
A.	2	4	4	1	3	2	2	1	3	1	1	1	2	3	1

HINT & SOLUTION

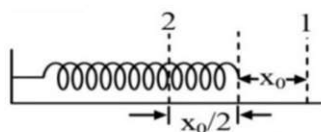
1. ANS - 1

$$F = ma = \frac{mv}{T} \quad (a = \frac{v-0}{T})$$

Instantaneous power

$$Fv = mav = \frac{mv}{T} at = \frac{mv}{T} \frac{v}{T} t = \frac{mv^2}{T^2} t$$

2. ANS - 3



Change in potential energy is independent of reference

$$U_2 - U_1 = \frac{1}{2} k \left(\frac{x_0}{2} \right)^2 - \frac{1}{2} k x_0^2 = -\frac{3}{8} k x_0^2$$

3. ANS - 4

From work energy theorems,

$$\frac{p^2}{2m} = \mu m g s \Rightarrow s = \frac{p^2}{2\mu m^2 g}$$

4. ANS - 1

Gain in KE = Loss in PE

5. ANS - 2

At the bottom most point, square of speed of bob 0

$$v^2 = 2gL(1 - \cos \alpha)$$

It will rise further to a height

$$h = \frac{v^2}{2g} = L(1 - \cos \alpha)$$

$$\text{or } (L - l)(1 - \cos \theta) = L(1 - \cos \alpha)$$

$$\therefore \theta = \cos^{-1} \left[\frac{L \cos \alpha - l}{L - l} \right]$$

6. ANS - 2

Power = Work / Time = Force × distance / Time = Force × velocity

$$\therefore \text{Force} \times \text{velocity} = \text{constant (K) or } (ma) \cdot (at) = K$$

$$\therefore s = \frac{1}{2} at^2 \therefore s = \frac{1}{2} (Kmt) \frac{1}{2t} = \frac{1}{2} (Km) \frac{1}{2t^{3/2}}$$

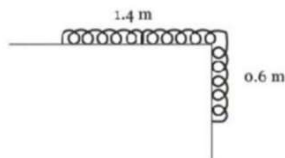
Or s is proportional to $t^{3/2}$

7. ANS - 2

In case of non-conservative forces, the work

done is dissipated as heat, sound, etc. ie, it does not increase the potential energy. But in case of conservative forces, work done is responsible for increasing or decreasing the potential energy.

8. ANS - 4



The center of mass of the hanging part is at 0.3 m from table

Mass of hanging part = $4 \times 0.6 = 1.2$ kg

So, $W = mgh = 1.2 \times 10 \times 0.3 = 3.6$ J

(center of hanged part distance from top is taken as several points are located at different distance from top)

9. ANS - 2

Since $K_1 > K_2$, therefore, P will develop greater restoring force than Q. So the applied force on P will be more than the applied force on Q. Extension is same in both the cases. So, more work is done on P.

10. ANS - 4

$$x = \frac{t^4}{4} + 3$$

$$\frac{dx}{dt} = \frac{4t^3}{4} + 0$$

$$V = t^3$$

Using work energy theorem

$$\begin{aligned} w = \Delta k &= \frac{1}{2} m(v(2))^2 - \frac{1}{2} m(v(0))^2 \\ &= \frac{1}{2} (2) [(2^3)^2 - 0] = 64 \text{ J} \end{aligned}$$

11. ANS - 1

$$U(x) = (x^2 - 3x) \text{ J}$$

For a conservative field, Force $F = - \frac{du}{dx}$

$$\therefore F = - \frac{d}{dx}(x^2 - 3x) = -(2x - 3) = -2x + 3$$

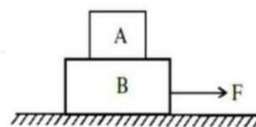
At equilibrium position, $F = 0$

$$\Rightarrow -2x + 3 = 0 \Rightarrow x = \frac{3}{2} \text{ m} = 1.5 \text{ m}$$

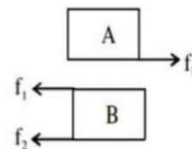
12. ANS - 1

Consider the blocks shown in the figure to be move together due to static friction between them.

The free body diagram of both the blocks are shown below.



Work done by static friction on A is positive and on B is negative



13. ANS - 2

Mass is displaced from (0,0) to (1,1). Therefore displacement is,

$$\vec{S} = (1 - 0)\hat{i} + (1 - 0)\hat{j}$$

$$\vec{S} = (\hat{i} + \hat{j})m$$

Given force

$$\vec{F} = (4\hat{i} + 4\hat{j}) \text{ N} ;$$

Work done by a constant force,

$$W = \vec{F} \cdot \vec{S}$$

$$W = (4\hat{i} + 4\hat{j}) \cdot (\hat{i} + \hat{j})$$

$$W = 4 + 4 = 8 \text{ J}$$

From work energy theorem - $\Delta K = W$

$$\frac{1}{2}m(v_2^2 - v_1^2) = W$$

$$\frac{1}{2}(1)(v_2^2 - 2^2) = 8$$

$$v_2^2 = 20$$

$$v_2 = 4.5 \text{ m/s}$$

14.ANS - 4

$$\text{Power} = \frac{\text{work}}{\text{time}} = \frac{\text{energy}}{\text{time}} = \frac{mgh}{t} = \frac{Wh}{t}$$

(W = weight)

$$\therefore \frac{P_1}{P_2} = \frac{W_1}{W_2} \times \frac{t_2}{t_1} = \frac{4}{3} \times \frac{11}{12} = \frac{11}{9}$$

(h is same in both cases)

15.ANS - 2

$$F = cx$$

$$W = \int F dx$$

$$W = \int_0^{x_1} cx dx$$

$$F = \left[c \frac{x^2}{2} \right]_0^{x_1}$$

$$F = c \frac{x_1^2}{2}$$

16.ANS - 2

According to Work Energy Theorem work done by all forces is equal to change in kinetic energy.

17.ANS - 3

We know that $U = ax + by$

$$\frac{\partial U}{\partial x} = \frac{\partial}{\partial x}(ax + by)$$

$$\frac{\partial U}{\partial x} = a$$

$$\frac{\partial U}{\partial y} = \frac{\partial}{\partial y}(ax + by)$$

$$\frac{\partial U}{\partial y} = b$$

$$\vec{F} = \left(\frac{\partial U}{\partial x} \hat{i} + \frac{\partial U}{\partial y} \hat{j} \right)$$

$$\vec{F} = -a\hat{i} - b\hat{j}$$

$$a = \frac{F}{m} = \frac{\sqrt{a^2 + b^2}}{m}$$

18.ANS - 2

Work done by both the persons is the same and is given as,

$W = mgh$, It does not depend upon the time taken to perform the work.

19.ANS - 3

Let work done by all forces other than conservative forces = W_{others}

Work done by conservative forces, W_{con} is negative of change in potential energy.

$W_{\text{con}} = -\Delta U$, From work energy theorem,

$$W_{\text{by all forces}} = \Delta K$$

$$W_{\text{others}} + W_{\text{con}} = \Delta K$$

$$W_{\text{others}} - \Delta U = \Delta K$$

$$W_{\text{others}} = \Delta U + \Delta K$$

Therefore, work done by all forces other than conservative forces is equal to change in total mechanical energy.

20.ANS - 1

From work energy theorem

$$W = \Delta K$$

$$\int_0^2 F dx = \frac{1}{2}m(v^2 - u^2)$$

$$\int_0^2 (3x^2 - 12x) dx = \frac{1}{2} 2(v^2 - (4\sqrt{5})^2)$$

$$[x^3 - 6x^2]_0^2 = v^2 - 80$$

$$8 - 24 + 80 = v^2$$

$$v = 8 \text{ ms}^{-1}$$

21. ANS - 1

Mass of the body (m) = 0.5 kg

Velocity of the body (v) = $ax^{\frac{3}{2}}$

Where $a = \frac{5m^{-\frac{1}{2}}}{s}$

Velocity of the body at x=0, $V_1 = 5 \times 0 = 0$

Velocity of the body at x=2, $V_2 = 5 \times \left(2^{\frac{3}{2}}\right) m/s$

According to work energy theorem,

Work done = change in K. E. = $\frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$

$$= \frac{1}{2}m(v_2^2 - v_1^2) = \frac{1}{2} \times 0.5 \left[\left(5 \times 2^{\frac{3}{2}}\right)^2 - 0^2 \right]$$

$$= \frac{1}{2} \times 0.5 \times 25 \times 8 = 50 \text{ J}$$

22. ANS - 2

In the potential energy curve, the point at maximum (crest) is the point of unstable equilibrium since, the displacement of particle towards the left or right will experience the increase or decrease in the force and potential energy, both are related by the relation,

$F = -\frac{\partial U}{\partial x}$, Hence, the point at unstable equilibrium is point B

23. ANS - 3

The spring will exert maximum force when the ball is at its lowest position. If the ball has descended through a distance x to reach the position.

$$mgx = \frac{1}{2}kx^2 \text{ or } x = \frac{2m}{k} \dots\dots(i)$$

For block B to leave the contact spring force

$$kx = Mg, \dots\dots(ii)$$

Comparing both the equation $m = M/2$

24. ANS - 1

$$\frac{dv}{dt} \propto -x \Rightarrow \frac{dv}{dx} \cdot \frac{dx}{dt} = -cx$$

$$Vdv = -cx dx$$

$$\frac{v^2}{2} = -\frac{cx^2}{2} \quad \text{kinetic energy} = \frac{mv^2}{2}$$

$$\text{kinetic energy} = \frac{mcx^2}{2}$$

$$k \propto x^2$$

25. ANS - 1

Nail has fallen through a total vertical distance of (h + x) hence loss in its

Potential energy = mg(h + x). If average retarding force exerted by sand on the nail is F, then work done - W = -Fx

For equilibrium, mg(h + x) = Fx

$$F = mg \frac{h+x}{x} = mg \left(\frac{h}{x} + 1 \right)$$

26. ANS - 2

$$W = \int_{(0,0)}^{(1,1)} \vec{F} \cdot d\vec{x}$$

$$\text{Here } d\vec{s} = dx\hat{i} + dy\hat{j} + dz\hat{k}$$

$$W = \int_{(0,0)}^{(1,1)} (x^2 dy + y dx) = \int_{(0,0)}^{(1,1)} (x^2 dy + x dx)$$

as (x = y)

$$W = \left[\frac{y^3}{3} + \frac{x^2}{2} \right]_{(0,0)}^{(1,1)} = \frac{5}{6} \text{ J}$$

27. ANS - 3

Loss in P.E. = gain in K.E. + work done against friction

$$mgR = \frac{1}{2}m(1.4gR) + W_f$$

$$W_f = 0.3mgR$$

$$\text{Now } W_{B \rightarrow C} = mgR + 0.3mgR = 1.3mgR$$

28. ANS - 2

Only external force can change the momentum of the system while internal forces can change the momentum of individual particles and hence the kinetic energy of the system.

29. ANS - 4

For the upper half

$$v^2 = u^2 + 2al/2 = 2(g\sin\theta)l/2 = gl\sin\theta$$

$$\Rightarrow 0 = u^2 + 2g(\sin\theta - \mu\cos\theta)\frac{l}{2}$$

$$\Rightarrow -gl\sin\theta = gl(\sin\theta - \mu\cos\theta)$$

$$\Rightarrow \mu\cos\theta = 2\sin\theta \Rightarrow \mu = 2\tan\theta$$

30. ANS - 2

Loss of energy = 35%

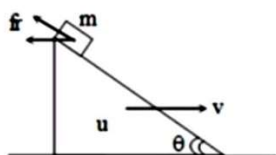
Remaining energy = 65%

$\Rightarrow$ New energy = 65% of initial energy

$$Mgh = \frac{65}{100}Mgh'$$

$$h' = \frac{65}{100} \times 100 = 65 \text{ m}$$

31. ANS - 2



As the block is rest with respect to the wedge

$$f_r = mg\sin\theta$$

Work done by friction

$$W = FS\cos\theta$$

$$W = -mg\sin\theta(vt)\cos\theta$$

$$W = -\frac{mg\sin 2\theta}{2}vt$$

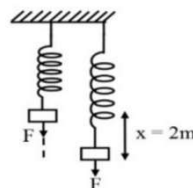
32. ANS - 4

$$W = -(Mg\sin\theta)d$$

33. ANS - 4

K.E is a scalar. Since it is zero, every individual velocity is zero. Momentum is a vector. If it is zero, it does not imply every velocity of particles in the system is equal to zero

34. ANS - 1



At equilibrium, $mg = kx_0$

$$\Rightarrow x_0 = \frac{mg}{k} = 1 \text{ m}$$

$$\therefore W = U_2 - U_1$$

$$= \frac{1}{2}kx_2^2 - \left[\frac{1}{2}kx_1^2 + mgh \right]$$

$$= \frac{1}{2}k(x_2^2 - x_1^2) - mgh$$

$$= \frac{1}{2} \times 100 \times (3^2 - 1^2) - 10 \times 10 \times 2 = 200 \text{ J}$$

35. ANS - 3

Potential energy + Kinetic energy
= Translation energy

Translation energy = 2 x Potential energy

$$T.E_1 = 4T.E \quad (\because V_1 = 2V)$$

$$\frac{PE^1}{KE^1} = \frac{PE}{8PE - PE} = \frac{1}{7}$$

36.ANS - 2

Tension in the string

$$T = M(g - a) = M\left(g - \frac{g}{2}\right) = \frac{Mg}{2}$$

$W = \text{Force} \times \text{displacement}$

$$= -\frac{Mgh}{2}$$

37.ANS - 2

When work is done upon the system by a conservative force then its potential energy increases

38.ANS - 1

$\frac{3}{4}$ th energy is lost i.e. $\frac{1}{4}$ th kinetic energy is left. Hence, its velocity becomes $\frac{v_0}{2}$ under a retardation of μg in time t_0

$\therefore$ from the first equation of motion

$$\frac{v_0}{2} = v_0 - \mu g t_0$$

$$\text{Or } \mu g t_0 = \frac{v_0}{2}$$

$$\text{Or } \mu = v_0 / 2 g t_0$$

39.ANS - 3

In a conservative field $F = -\frac{dU}{dr}$

$$F = -\frac{d}{dx}(ax^2 - bx) = b - 2ax$$

40.ANS - 1

$$p = mg \sin \theta \times V = 4000 \text{ W}$$

41.ANS - 1

From work energy theorem for upward motion

$$\frac{1}{2}m(16)^2 = mgh + W$$

For downward motion,

$$\frac{1}{2}m(8)^2 = mgh - W \Rightarrow h = 8 \text{ m}$$

42.ANS - 1

Energy stored in a spring is given by

$$U = \frac{1}{2}kx^2$$

Where k is spring constant, Now applying Hooke's law, we have

$$\text{Tension } T = kx \Rightarrow \text{Extension } x = T/k \Rightarrow U = \frac{T^2}{2k}$$

43.ANS - 2

$$v = \frac{dx}{dt} = \frac{1}{4} \times 4t^3 = t^3 = 1 \text{ ms}^{-1}$$

work done = change in K.E.

$$= \frac{1}{2} \times 1 \times 1 - \frac{1}{2} \times 1 \times 0 = \frac{1}{2} \text{ J}$$

44.ANS - 3

$$P = Fv \quad \text{Or } P = \left(mv \frac{dv}{dx}\right) V$$

$$\text{or } \int_0^V v^2 dv = \int_0^x \frac{P}{m} dx$$

$$\frac{v^3}{3} = \frac{Px}{m} \quad \text{or } v = \left(\frac{3Px}{m}\right)^{1/3}$$

45.ANS - 1

Weight of train $10^7 \text{ N} = 10^6 \text{ kgf}$

$$\text{Frictional force, } F = \frac{0.5}{100} \times 10^6 = 5000 \text{ kgf}$$

$$F = 50000 \text{ N}$$

$$v = 36 \text{ km/h} = 10 \text{ m/s}$$

$$P = F \times v = 50000 \times 10$$

$$= 5 \times 10^5 \text{ W} = 500 \text{ kW}$$

TOPIC : CENTRE OF MASS & COLLISIONS

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	3	1	2	3	2	3	1	1	4	3	3	3	3	3	1
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
A.	2	3	1	2	3	3	1	2	1	2	1	3	3	1	4
Q.	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	2	1	1	2	1	2	2	3	2	1	1	4	1	1	2

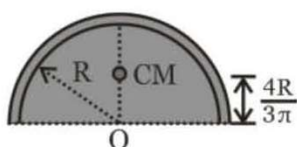
HINT & SOLUTION

46. ANS – 3

$$e = \frac{\text{velocity of separation}}{\text{velocity of approach}}$$

$$e = \sqrt{\frac{h_2}{h_1}} = \sqrt{\frac{81}{100}} = 0.9$$

47. ANS – 1



Centre of mass of uniform semicircular disc of radius R lies at a distance equal to $\frac{4R}{3\pi}$ from the centre of the axis of symmetry as shown in figure

48. ANS – 2

$$x_{com} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$

$$1 = \frac{(10 \times 7) + (30x)}{40}$$

$$x = -1 \text{ cm}$$

49. ANS – 3

Let v be the velocity after collision, collision is perfectly elastic $e = 1$

Or relative velocity of separation = relative velocity of approach

$$v - 1 = 2 + 1 \quad \text{Or} \quad v = 4 \text{ m/s}$$

50. ANS – 2

For Block A to be in equilibrium, the spring force acting on it should be equal to its weight. This means $F_{spring} = 2mg$

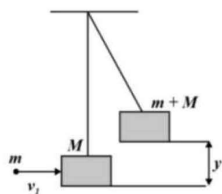
$$a = \frac{3mg - 2mg}{3m} = \frac{g}{3}$$

51. ANS – 3

$$X_{cm} = \frac{\int x dm}{\int dm}$$

$$X_{cm} = \frac{\int x \lambda dx}{\int \lambda dx} = \frac{\int_0^l x k x^2 dx}{\int_0^l k x^2 dx} = \frac{\frac{l^4}{4}}{\frac{l^3}{3}} = \frac{3}{4} l$$

52. ANS - 1



As blocks rises to a height y , therefore common velocity of blocks just after the collision is

$$v_c = \sqrt{2gy}$$

As the collision is perfectly inelastic, using conservation of linear momentum,

$$mv_1 = (m + M)\sqrt{2gy}$$

$$\therefore v_1 = \frac{m + M}{m}\sqrt{2gy}$$

53. ANS - 1

For elastic collision,

$$v_2 - v_1 = u_1 - u_2 \text{---(i)}$$

Momentum conservation,

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

For equal masses,

$$u_1 + u_2 = v_1 + v_2 \text{---(ii)}$$

Solving (i) & (ii) we get,

$$v_1 = u_2 \text{ and } v_2 = u_1$$

54. ANS - 4

A shell fired at an angle ' θ ', velocity of the shell at the highest point is $u \cos \theta$.

Let its mass be m . The speed of one piece is $u \cos \theta$ and let the speed of other piece be v .

Hence at the highest point, from conservation of momentum

$$m u \cos \theta = -\frac{m}{2} u \cos \theta + \frac{m}{2} v$$

$$m u \cos \theta + \frac{m}{2} u \cos \theta = \frac{m}{2} v$$

$$\frac{2m u \cos \theta + m u \cos \theta}{2} = \frac{m}{2} v$$

$$mv = 3m u \cos \theta$$

$$v = 3u \cos \theta \text{ ... (i)}$$

The kinetic energy

$$E_1 = \frac{1}{2} \times \frac{m}{2} u^2 \cos^2 \theta$$

$$E_1 = \frac{1}{4} m u^2 \cos^2 \theta \text{ ... (ii)}$$

Similarly

$$E_2 = \frac{1}{2} \times \frac{m}{2} \times 9u^2 \cos^2 \theta$$

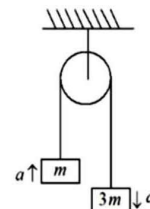
$$E_2 = \frac{9}{4} m u^2 \cos^2 \theta \text{ ... (iii)}$$

The relation between E_1 and E_2

$$E_2 = 9 \times E_1 \text{ (from the equation (ii))}$$

$$E_2 = 9E_1$$

55. ANS - 3



The magnitude of the acceleration of each mass with respect to the pulley

$$a = \frac{3mg - mg}{3m + m} = \frac{g}{2}$$

Acceleration of center of mass

$$= \frac{3m(g/2) - m(g/2)}{3m + m} = \frac{g}{4}$$

56. ANS - 3

If the height of the stone is h_1 and h_2 after $t=3s$, Then

$$h_1 = 100 - \frac{1}{2}gt^2 = 55 \text{ m}$$

$$h_2 = 100 - \frac{1}{2}gt^2 = 55 \text{ m}$$

$$h_{cm} = \frac{m_1 h_1 + m_2 h_2}{m_1 + m_2} = \frac{m(55+5)}{2m} = 55 \text{ m}$$

57. ANS - 3

In collision,

final momentum = initial momentum

$$\vec{p}_f = \vec{p}_i$$

$$\vec{p}_2 + (1)(0.3)\hat{j} = (1)(0.4)\hat{i}$$

$$\vec{p}_2 = 0.4\hat{i} - 0.3\hat{j}$$

$$\vec{p}_2 = 0.5\text{kg} - \text{m/s}$$

58. ANS - 3

Let $\vec{v}$ be the velocity of the third part. Therefore, from the conservation of momentum,

$$0 = 1(4\hat{i}) + 1(2\hat{j}) + 1\vec{v}$$

$$\therefore \vec{v} = -4\hat{i} - 2\hat{j}$$

The energy released in the explosion is the sum of the kinetic energy of all three parts,

$$\therefore E = \frac{1}{2}(1)(4)^2 + \frac{1}{2}(1)(2)^2 + \frac{1}{2}(1)(4^2 + 2^2)$$

$$E = 20 \text{ J}$$

59. ANS - 3

The trolley shall move backwards to conserve momentum. The backward momentum would be shared by both the trolley and the man. Applying conservation of momentum,

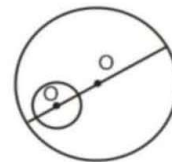
$$60 \times 1 = (240 + 60) v \quad \text{or} \quad 60 = 300v$$

$$\text{Or } v = \frac{60}{300} = \frac{1}{5} \text{ m/s} = 0.2 \text{ m/s}$$

$$\text{Speed of man w.r.t ground} = (1-0.2) \text{ m/s} = 0.8 \text{ m/s}$$

$$\text{Displacement of man} = 0.8 \times 4 \text{ m} = 3.2 \text{ m}$$

60. ANS - 1



For the calculation of the position of centre of mass, cut off mass is taken as negative. The mass of disc is

$$m_1 = \pi r_1^2 \sigma = \pi 6^2 \sigma = 36\pi\sigma$$

Where, σ = surface mass density. The mass of cut portion is

$$m_2 = \pi(1)^2 \sigma = \pi\sigma$$

$$x_{cm} = \frac{m_1 x_1 - m_2 x_2}{m_1 - m_2}$$

Taking origin at the centre of disc,

$$x_1 = 0, x_2 = 3 \text{ cm}$$

$$x_{cm} = \frac{36\pi\sigma \times 0 - \pi\sigma \times 3}{36\pi\sigma - \pi\sigma} = \frac{-3\pi\sigma}{35\pi\sigma} = -\frac{3}{35} \text{ cm}$$

61. ANS - 2

The centre of mass of the two parts B and C of the body will not shift. It is because after breaking into two parts no external force acting on the system it still under the effect of gravity. Hence the centre of mass of two-part B and C will follow the original path.

62. ANS - 3

Given, $m_1 = 6 \text{ kg}$, $m_2 = 4 \text{ kg}$

$$\vec{V}_1 = 5\hat{i} - 2\hat{j} + 10\hat{k}, \quad \vec{V}_2 = 10\hat{i} - 2\hat{j} + 5\hat{k}$$

The velocity of centre of mass is

$$\vec{V} = \frac{m_1 \vec{v}_1 + m_2 \vec{v}_2}{m_1 + m_2}$$

$$= \frac{6(5\hat{i} - 2\hat{j} + 10\hat{k}) + 4(10\hat{i} - 2\hat{j} + 5\hat{k})}{6 + 4}$$

$$= \frac{70\hat{i} - 20\hat{j} + 80\hat{k}}{10} = 7\hat{i} - 2\hat{j} + 8\hat{k}$$

63. ANS - 1

$$m_1 = 1 \text{ kg}, m_2 = 2 \text{ kg}, m_3 = 3 \text{ kg}$$

Position of centre of mass (2,2,2) $m_4 = 4 \text{ kg}$

New position of centre of mass (0,0,0)

For initial position

$$X_{\text{COM}} = \frac{m_1x_1 + m_2x_2 + m_3x_3}{m_1 + m_2 + m_3}$$

$$2 = \frac{m_1x_1 + m_2x_2 + m_3x_3}{1 + 2 + 3}$$

$$= m_1x_1 + m_2x_2 + m_3x_3 = 12$$

$$\text{similarly, } m_1y_1 + m_2y_2 + m_3y_3 = 12$$

for new position,

$$X_{\text{COM}} = \frac{m_1x_1 + m_2x_2 + m_3x_3 + m_4x_4}{m_1 + m_2 + m_3 + m_4}$$

$$0 = \frac{12 + 4 \times x_4}{1 + 2 + 3 + 4}$$

$$= 4x_4 = -12$$

$$x_4 = -3$$

$$\text{similarly } y_4 = -3$$

$$z_4 = -3$$

$$\therefore \text{position of fourth mass } (-3, -3, -3)$$

64. ANS - 2

Let the speed of ball be v and speed of the wedge be v_2 . Using conservation of momentum

$$0 = mv + 2mv_2$$

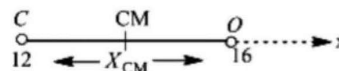
$$v_2 = -\frac{v}{2}$$

Now, using work-energy theorem on the system,

$$mgR = \frac{1}{2}mv^2 + \frac{1}{2}(2m) \times \left(\frac{v}{2}\right)^2$$

$$\Rightarrow V = \sqrt{\frac{4gR}{3}}$$

65. ANS - 3



$$X_{\text{com}} = \frac{m_1x_1 + m_2x_2}{m_1 + m_2}$$

$$\therefore X_{\text{com}} = \frac{(12 \times 0) + 16 \times 1.13}{12 + 16} = 0.6457 \text{ \AA}$$

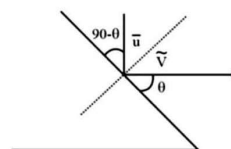
66. ANS - 3

If we consider a system consists of a gun plus mass m then no external force is present on the system. So, the acceleration of the centre of mass of the system will be zero. Since initially the system is in rest. So, the centre of mass will always remain at rest.

67. ANS - 1

If $M=M'$ then bullet will transfer the whole of its velocity (and consequently 100% of its KE) to b block and will itself come to rest as per the theory of collision.

68. ANS - 2



$$\vec{u} = -\sqrt{2gh}\hat{j}$$

$$|\vec{v}| = |\vec{u}|$$

$$\vec{v} = \sqrt{2gh}\hat{i}$$

$$u \cos \theta = v \sin \theta$$

$$\tan \theta = 1$$

$$\theta = 45^\circ$$

69. ANS - 1

Let m_1 and m_2 be the masses of the particles.

By principles of conservation of momentum,

$$m_1 u_1 + 0 = m_2 v - m_1 v \dots \dots \dots (i)$$

By Newton's law,

$$\therefore e = 1$$

$$2v = u_1$$

Substituting this in (i)

$$2m_1 v = m_2 v - m_1 v$$

$$2m_1 = m_2 - m_1$$

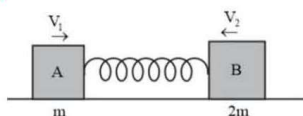
$$3m_1 = m_2$$

$$\frac{m_1}{m_2} = \frac{1}{3}$$

70. ANS - 2

A large number of particles are situated at a distance R from the origin. If particles are distributed and make a circular boundary around the origin, the center of mass will lie at the origin. Whereas, If the particles are not uniformly distributed, then distance can be any value between 0 and R . This implies the distance between the center of mass and the origin is always less than or equal to R .

71. ANS - 1



Relative velocity will be

$$v_1 + v_2$$

Using conservation of linear momentum

$$m v_1 = 2m v_2$$

$$\Rightarrow v_1 = 2v_2$$

Using conservation of energy

$$\frac{1}{2} kx^2 = \frac{1}{2} m v_1^2 + \frac{1}{2} 2m v_2^2$$

$$\frac{1}{2} kx^2 = \frac{1}{2} m (2v_2)^2 + m v_2^2$$

$$\frac{1}{2} kx^2 = 2m v_2^2 + m v_2^2$$

$$3m v_2^2 = \frac{kx^2}{2} \quad v_2^2 = \frac{kx^2}{6m}$$

$$v_2 = \sqrt{\frac{k}{6m}} x$$

$$\text{Relative velocity} = 3v_2 = 3 \sqrt{\frac{k}{6m}} x = \sqrt{\frac{3k}{2m}} x$$

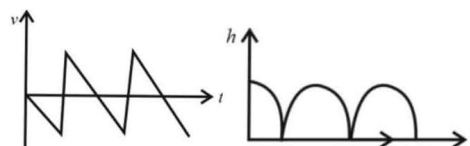
72. ANS - 3

Let the velocity of boat be V opposite to the man. By momentum conservation,

$$-40V + 80(6 - V) = 0, V = 4 \text{ m/s}$$

$$\text{Velocity of the man} = 6 - 4 = 2 \text{ m/s}$$

73. ANS - 3



When rubber ball strikes the elastic surface, then after collision its velocity will be opposite in direction without change in speed. Since h is positive, thus initial velocity will be negative. Hence option (3) is correct.

74. ANS - 1

Since no external force is present on the rod in horizontal direction, the velocity of the centre of mass of rod, in horizontal direction, remains constant i.e., zero. So, centre of mass of rod drops vertically downwards.

75. ANS - 4

$$V_c = \frac{2mv - mv}{(2m + m)} = \frac{v}{3}$$

$$V_{rel} = v - \frac{v}{3} = \left(\frac{2v}{3}\right)$$

$$r_1 = \frac{mL}{3m} = \frac{L}{3}$$

$$\therefore T = \frac{(2m) \times \left(\frac{2v}{3}\right)^2}{\frac{L}{3}} = \frac{8mv^2}{3L}$$

76. **ANS - 2**

The velocity of the centre of mass changes only if some external force (net external resultant) acts on it. Internal forces do not affect the motion of the centre of mass.

77. **ANS - 1**

The velocity of wedge = $-\frac{mv \cos \theta}{m+M}$

θ : Angle made by $\vec{v}_{man}$ with the horizontal

78. **ANS - 1**

Let v_1 = speed of second mass after collision
momentum is conserved

Along x-axis $mv_1 \cos \theta = mv$

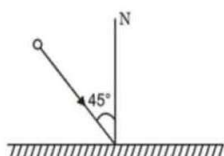
Along y-axis $mv_1 \sin \theta = mv/\sqrt{3}$

From (i) and (ii)

$$\therefore (mv_1 \cos \theta)^2 + (mv_1 \sin \theta)^2 = (mv)^2 + \left(\frac{mv}{\sqrt{3}}\right)^2$$

$$\text{or } m^2 v_1^2 = \frac{4m^2 v^2}{3} \text{ or } v_1 = \frac{2}{\sqrt{3}} v$$

79. **ANS - 2**

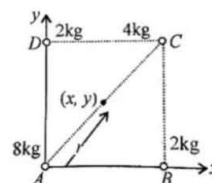


Since surface is smooth so force is normal so horizontal component will remain unchanged while vertical component decreases or remain same ($e < 1$) that's why angle from vertical will increase or remains same.

80. **ANS - 1**

Balls are identical and velocity exchanges if elastic head on collision occurs.

81. **ANS - 2**



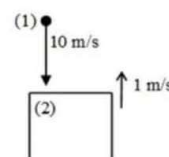
According to figure let A is the origin and co-ordinates of centre of mass be (x, y) , then,

$$x = \frac{m_1 x_1 + m_2 x_2 + m_3 x_3 + m_4 x_4}{m_1 + m_2 + m_3 + m_4}$$

$$= \frac{0 + 2 \times \frac{80}{\sqrt{2}} + 4 \times \frac{80}{\sqrt{2}} + 0}{16} = \frac{30}{\sqrt{2}}$$

Similarly, $y = \frac{30}{\sqrt{2}}$ so, $r = \sqrt{x^2 + y^2} = 30 \text{ cm}$

82. **ANS - 2**



Velocity after a fall of 5 m is -

$$u_1 = \sqrt{2 \times 10 \times 5} = 10 \text{ ms}^{-1}$$

$$\bar{v}_1 - \bar{v}_2 = e(\bar{u}_2 - \bar{u}_1)$$

$$v_1 - 1 = 1 - (-10)$$

$$v_1 = 12 \text{ ms}^{-1}$$

83. **ANS - 3**

Velocity of the colliding body after elastic collision is,

$$mv_1 + 0 = m \frac{v_1}{2} + m'v_2$$

$$e = 1 = \frac{\left(v_2 - \frac{v_1}{2}\right)}{v_1}$$

$$\Rightarrow m' = \frac{m}{3} \text{ kg}$$

84. ANS - 2

If external forces acting on a system have zero resultant, then acceleration is zero and velocity remains constant

85. ANS - 1

$$u = (3\hat{i} + \hat{j})$$

Component of velocity parallel to wall i.e. y-component remains unchanged Also, component of velocity perpendicular to the wall becomes e times

$$\vec{v}_\perp = -e\vec{u}_\perp$$

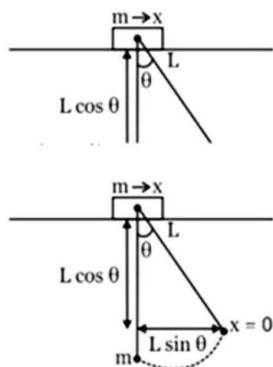
$$\vec{v}_\perp = -\frac{1}{2}(3\hat{i})$$

$$\therefore \text{Impulse } \vec{J} = m(\vec{v} - \vec{u})$$

$$\vec{J} = m(\vec{v}_\perp - \vec{u}_\perp)$$

$$\vec{J} = m\left[\left(-\frac{3}{2}\hat{i}\right) - (3\hat{i})\right] = -\frac{9}{2}m\hat{i}$$

86. ANS - 1



$$\Delta x_{cm} = 0$$

$$0 = -(L \sin \theta - x)m + mx$$

$$L \sin \theta - x = x$$

$$2x = L \sin \theta$$

$$x = \frac{L \sin \theta}{2}$$

87. ANS - 4

$$m_G v_G = m_B v_B \Rightarrow v_B = \frac{m_G v_G}{m_B} = \frac{1 \times 5}{10 \times 10^{-3}}$$

$$= 500 \frac{M}{S}$$

88. ANS - 1

$$\text{Initial height of centre of mass } h_i = \frac{a}{2}$$

$$\text{As the volume remains same } 8l^3 = a^3$$

$$l = \frac{a}{2}$$

Final length of the centre of mass

$$h_f = \frac{8l}{2} = 4l$$

$$\therefore \Delta h = 4l - \frac{a}{2} = 2a - \frac{a}{2} = \frac{3a}{2}$$

89. ANS - 1

After each collision speed becomes e times

Speed after 1st collision is $v_1 = ev_0$

$$h_1 = \frac{v_1^2}{2g} = e^2 \left(\frac{v_0^2}{2g} \right) = e^2 h$$

Similarly maximum height after 2nd collision is,

$$h_2 = e^2 h_1 = e^4 h$$

Maximum height after 3rd collision is,

$$h_3 = e^2 h_2 = e^6 h$$

So, the height to which the ball goes up after it rebounds for the nth time is $h_n = e^{2n} h$

90. ANS - 2

$$mu = Mv$$

$$\Rightarrow \frac{m}{M} = \frac{v}{u}$$

$$\text{but } e = \frac{v}{u} \leq 1 \Rightarrow e = \frac{m}{M} \leq 1$$

TOPIC : CIRCULAR MOTION

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
A.	1	1	2	1	3	2	1	2	1	3	1	2	3	1	3
Q.	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	4	3	3	3	1	1	2	4	2	4	3	2	2	4	3
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135
A.	1	2	1	4	4	2	2	4	1	1	2	4	3	4	1

HINT & SOLUTION

91. **ANS - 1**

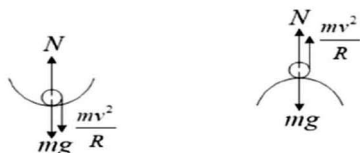
The tangential acceleration of the particle is

$$a_t = \frac{dV}{dt} = \frac{4.0 - 2.0}{4} = 0.5 \text{ m/s}^2$$

The angular acceleration is

$$\alpha = \frac{a_t}{r} = \frac{0.5}{0.5} = 1 \frac{\text{rad}}{\text{s}^2}$$

92. **ANS - 1**



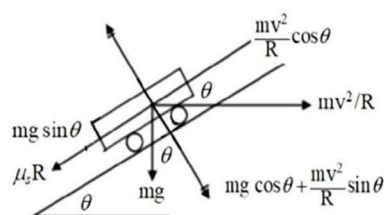
$$N_A = mg + \frac{mv^2}{R_A}$$

$$N_B = mg - \frac{mv^2}{R_B}$$

$$N_C = mg + \frac{mv^2}{R_C}$$

From the curve, $R_A < R_C$, $N_A > N_C$,
 N_A is maximum

93. **ANS - 2**



$$N = mg \cos \theta + \frac{mv^2}{R} \sin \theta \text{ and}$$

$$f_{\max} = \mu_s N = \mu_s mg \cos \theta + \frac{\mu_s mv^2}{R} \sin \theta$$

$$\text{Now } mg \sin \theta + f_{\max} = \frac{mv^2}{R} \cos \theta$$

$$mg \sin \theta + \mu_s mg \cos \theta + \frac{\mu_s mv^2}{R} \sin \theta = \frac{mv^2}{R} \cos \theta$$

$$\text{on solving } V = \sqrt{gR \frac{\mu_s + \tan \theta}{1 - \mu_s \tan \theta}}$$

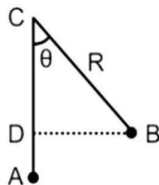
94. **ANS - 1**

Tension in the string

$$\text{Initial } T = m(\omega^2) r$$

$$\text{Finally, } T' = m(r/2) (2\omega)^2 = 2mr\omega^2 \\ = 2T$$

95. **ANS - 3**



$$V = \sqrt{4gR} = 2\sqrt{gR}$$

$$u = \frac{\sqrt{4gR}}{2} = \sqrt{gR}$$

Conservation of mechanical energy

$$\frac{1}{2}mv^2 = \frac{1}{2}mu^2 + mgh$$

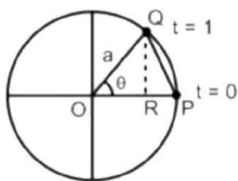
$$\frac{1}{2}m(4gR) = \frac{1}{2}m(gR) + mgR(1 - \cos \theta)$$

Cancelling the common term mgR we have

$$2 = \frac{1}{2} + 1 - \cos \theta$$

$$\cos \theta = -1/2$$

96. **ANS - 2**



In time t the particle has rotated through an angle $\theta = \omega t$

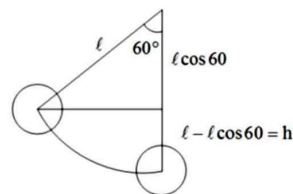
This means the magnitude of the displacement of the particle is

$$s = PQ = \sqrt{QR^2 + PR^2}$$

$$= \sqrt{(a \sin \omega t)^2 + (a - a \cos \omega t)^2}$$

$$= 2a \sin \left(\frac{\omega t}{2} \right)$$

97. **ANS - 1**



This means $h = l(1 - \cos 60^\circ)$

$$\text{Or } h = l \times \frac{1}{2}$$

$$\text{Velocity of ball} = \sqrt{2gh} \Rightarrow v = \sqrt{gl}$$

98. **ANS - 2**

Let x be increase in length of the spring. The particle would move in a circular path of radius $(l + x)$.

Centripetal force = force due to the spring

$$m(l + x)\omega^2 = kx$$

$$\therefore x = \frac{m\omega^2 l}{k - m\omega^2}$$

99. **ANS - 1**

For constant speed along circular path acceleration is centripetal which is constant in magnitude $a_c = \frac{v^2}{r}$ and always directed towards the center so the direction changes continuously.

100. **ANS - 3**

Initially speed of ball B is less hence centrifugal in non-inertial frame is less, so ball is in contact with the inner surface when speed will be sufficient, ball will lose contact with inner surface and goes in contact with the outer surface.

101. **ANS - 1**

Maximum speed

$$v = \sqrt{\mu rg} = \sqrt{0.4 \times 30 \times 9.8} = 10.84 \text{ m/s}$$

102. **ANS - 2**

The acceleration vector acts along the radius of the circle, given statement is false.

103. **ANS - 3**

$$\tan \theta = \frac{v^2}{rg} \Rightarrow r \propto v^2$$

$$\frac{r_2}{r_1} = \left(\frac{v_2}{v_1}\right)^2$$

$$\text{or } \frac{r_2}{r_1} = \left(\frac{1.1v}{v}\right)^2 \Rightarrow r_2 = 24.2 \text{ m}$$

104. **ANS - 1**

$$\tan \theta = \frac{v^2}{2g}, v = \sqrt{rg \tan \theta} = \sqrt{\frac{rgh}{b}}$$

where h is the elevation of the other end of the road and b is the width of the road

$$v = \sqrt{1.5 \times \frac{50}{100} \times 10} = 8.5 \text{ m/s}$$

105. **ANS - 3**

$$\text{At } 45^\circ, a_t = a_r = 9$$

$$\frac{v^2}{r} = \frac{v^2}{4} = 9$$

$$v = 6 \text{ ms}^{-1}$$

$$6 = 0 + 9t$$

$$t = \frac{2}{3} \text{ s}$$

106. **ANS - 4**

Given, centripetal acceleration $= 4/r^2$

$$\therefore \frac{v^2}{r} = \frac{4}{r^2}$$

$$\text{or } v = \frac{2}{\sqrt{r}}$$

$$\text{hence, } p = \frac{2m}{\sqrt{r}}$$

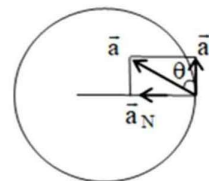
107. **ANS - 3**

The coin performs circular motion in a circle of radius r. It will fly off when centrifugal force is greater than the centripetal force provided by friction, Hence, at the moment when the coin is about to fly off,

$$\mu N = m\omega^2 r \text{ or } \mu mg = m\omega^2 r$$

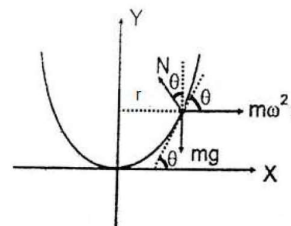
$$\omega = \sqrt{\frac{\mu g}{r}}$$

108. **ANS - 3**



$$\tan \theta = \frac{a_N}{a_t}$$

109. **ANS - 3**



$$N \cos \theta = mg$$

$$\text{And } N \sin \theta = m\omega^2 r$$

$$\therefore \tan \theta = \frac{\omega^2 r}{g} \dots (i)$$

$$\text{Given } y = x^2$$

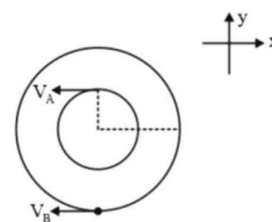
$$\therefore \frac{dy}{dx} = 2x$$

$$\text{Or } \tan \theta = 2 \times 1 = 2 \dots (ii)$$

From the above equation we get,

$$\omega = \sqrt{2g} \quad (r = 1 \text{ m})$$

110. **ANS - 1**

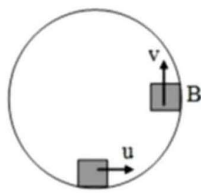


$$t = \frac{\pi}{2\omega}$$

$$\vec{V}_A = -\omega R_1 \hat{i}$$

$$\vec{V}_B = -\omega R_2 \hat{i}$$

111. **ANS - 1**



Speed at lowest point, to just complete the circle

$$u^2 = 5gR$$

Speed when velocity becomes vertical, is

$$v^2 = u^2 - 2gR$$

$$v^2 = 5gR - 2gR = 3gR$$

Tangential acceleration at B is

$$a_t = g \text{ (downwards)}$$

centripetal acceleration at B is

$$a_c = \frac{v^2}{R} = 3g$$

Total acceleration will be,

$$a = \sqrt{a_c^2 + a_t^2} = g\sqrt{10}$$

112. **ANS - 2**

$$T - mg \cos \theta = \frac{mv^2}{l}$$

$$\text{And } v^2 - u^2 = -2gl(1 - \cos \theta), \text{ Also } T = mg$$

113. **ANS - 4**

$$a_r \propto t; \frac{v^2}{r} = kt$$

$$v^2 = krt; 2v \frac{dv}{dt} = kr$$

$$a_t = \frac{kr}{2v} = \frac{kr}{2\sqrt{rkt}} = \frac{1}{2} \sqrt{\frac{kr}{t}}$$

$$a_t^2 a_r = \frac{1}{4} \frac{kr}{t} \times kt = \text{independent of } t$$

114. **ANS - 2**

$$\omega_{av} = \frac{\Delta \theta}{\Delta t}$$

$$= \frac{[10 - 5(3) + 4(3)^2] - [10 - 5(1) + 4(1)^2]}{3 - 1}$$

$$\omega_{av} = 11 \text{ rads}^{-1}$$

115. **ANS - 4**

Speed is constant but the direction is changing continuously hence velocity is changing, acceleration is acting radially inward, hence acceleration is also changing.

116. **ANS - 3**

$$r = 0.25 \text{ m}$$

$$\omega = 2 \frac{\text{rev}}{\text{s}}$$

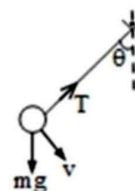
$$\omega = 2 \times 2\pi \text{ rad/s}$$

$$\text{We know that } a_c = \omega^2 r$$

$$\Rightarrow a_c = (2 \times 2\pi)^2 \times 0.25$$

$$a_c = 4\pi^2 \text{ m/s}^2$$

117. **ANS - 2**



FBD of body at the position shown in diagram

$$T - mg \cos \theta = \frac{mv^2}{l} \dots \dots \dots (i)$$

Applying work energy theorem

$$\Rightarrow mgl \cos \theta = \frac{1}{2} mv^2 - \frac{1}{2} m v_0^2 \dots \dots \dots (ii)$$

$$T = 2mg \dots \dots \dots (iii)$$

$$\text{By solving } \cos \theta = \frac{1}{4}$$

118. **ANS - 2**

While crossing the equilibrium position

$$T = mg + \frac{mv^2}{r} = mg + \frac{m}{l} [2gl(1 - \cos \theta)]$$

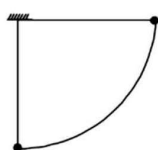
$$= mg + 2mg - 2mg \cos \theta$$

$$= 3 - 2 \cos \theta = \frac{2940}{(150)(9.8)} = 2$$

$$2 \cos \theta = 1$$

$$\cos \theta = \frac{1}{2} \rightarrow \theta = 60^\circ$$

119. **ANS - 4**



$$V = \sqrt{2gh} \text{ when 100\% energy is converted}$$

$$\text{But here } \frac{1}{2}mv^2 = 0.9mgh$$

$$v = \sqrt{1.8gh} = \sqrt{1.8 \times 10 \times 2} = 6 \text{ m/s}$$

120. **ANS - 3**

121. **ANS - 1**

Centrifugal force does not depend upon angular velocity of particle. It depends upon angular velocity of frame of reference.

$$\text{Centrifugal force} = F = m\omega^2 r$$

122. **ANS - 2**

$$F = \frac{mv^2}{r} \rightarrow F \propto v^2$$

If v becomes double then F (tendency to overturn) will become four times.

123. **ANS - 1**

$$\text{Tension at the top point } T + mg = \frac{mv^2}{r}$$

$$\text{So, } T = \frac{1 \times 4^2}{1} - 1 \times 10 = 6 \text{ N}$$

124. **ANS - 4**

$$\text{Given- } v = 20 \text{ m/s}$$

$$R = 100 \text{ m}$$

$$\frac{dv}{dt} = 3 \text{ m/s}^2$$

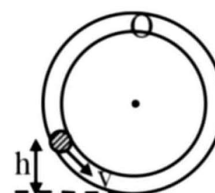
$$a_c = \frac{v^2}{R} = \frac{(20)^2}{100} = 4 \text{ m/s}^2$$

$$a_t = \frac{dv}{dt} = 3 \text{ m/s}^2$$

$$a = \sqrt{a_c^2 + a_t^2} = \sqrt{4^2 + 3^2} = 5 \text{ m/s}^2$$

125. **ANS - 4**

To reach at highest point of tube.



$$\text{Final velocity, } v = 0$$

$$\therefore \frac{1}{2}mv^2 + mgh = mg(2R)$$

$$\Rightarrow v = \sqrt{2g(2R - h)}$$

126. **ANS - 2**

$$\text{Critical velocity at top most point } v = \sqrt{rg}$$

Velocity when string becomes horizontal

$$v' = \sqrt{v^2 + 2gr}$$

$$= \sqrt{3gr}$$

$$\text{Centripetal acceleration} = \frac{v'^2}{r} = \frac{3rg}{r} = 3g$$

127. **ANS - 2**

$$\text{Given, } f = 1200 \text{ rpm,}$$

$$r = 30\text{cm} = \frac{30}{100}\text{m}$$

Acceleration of a point at the tip of the blade
= centripetal acceleration

$$= \omega^2 r = (2\pi f)^2 r$$

$$= \left(2 \times \frac{22}{7} \times \frac{1200}{60}\right)^2 \times \frac{30}{100} = 4740\text{ms}^{-2}$$

128. **ANS - 4**

$$\text{Kinetic energy} = \frac{1}{2}mv^2 = K = as^2$$

$$\text{or } mv^2 = 2as^2$$

$$\text{Centripetal force} = \frac{mv^2}{R} = \frac{2as^2}{R}$$

129. **ANS - 1**

$$f_s \leq f_{s(\text{max})}$$

Centripetal force is equal to maximum static friction

$$\frac{mv^2}{r} \leq \mu mg$$

$$\mu \geq \frac{v^2}{rg}$$

$$\mu_{\min} = \frac{v^2}{rg}$$

$$\mu_{\min} = \frac{(26)^2}{92 \times 9.8}$$

$$\mu_{\min} = 0.75$$

130. **ANS - 1**

Let particle is at angular position θ then
distance travelled = l

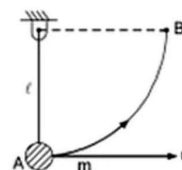
$$\text{But } a_t = \frac{dv}{dt} = \frac{a}{2\sqrt{l}} \frac{dl}{dt} = \frac{av}{2\sqrt{l}} = \frac{a^2}{2}$$

$$\text{And } a_c = \frac{v^2}{r} = \frac{a^2 l}{r}$$

Angle between a_{net} and a_t

$$\alpha = \tan^{-1} \left(\frac{2l}{r} \right)$$

131. **ANS - 2**



$$\text{Kinetic energy at A} = \frac{1}{2}mu^2$$

$$\text{Potential energy at B} = mgh$$

For conservation of energy,

$$\frac{1}{2}mu^2 = mgl$$

$$u = \sqrt{2gl}$$

132. **ANS - 4**

$$\omega_0 = 0, \omega = 24 \text{ rad s}^{-1}, t = 8\text{s}$$

$$\therefore \alpha = \frac{\omega - \omega_0}{t} = \frac{24}{8} = 3 \text{ rads}^{-2}$$

$$\theta = \omega_0 t + \frac{1}{2}\alpha t^2 = 0 + \frac{1}{2} \times 3 \times (8)^2$$

$$= \frac{3 \times 64}{2} = 96 \text{ rad.}$$

133. **ANS - 3**

Because of uniform angular acceleration.
We can use the equation of motion in rotation

$$\theta = \frac{1}{2}\alpha(2)^2 [\omega_0 = 0]$$

$$\theta = 2\alpha$$

And

$$\theta + \theta' = \frac{1}{2}\alpha(2+2)^2 [\omega_0 = 0]$$

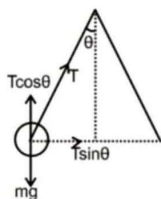
$$\theta + \theta' = 8\alpha$$

$$\text{Hence } \theta' = 3\theta$$

134. **ANS – 4**

$$T \sin \theta = m\omega^2 r$$

$$T \sin \theta = m\omega^2 L \sin \theta$$



$$\Rightarrow T = m\omega^2 L$$

$$\Rightarrow 324 = \frac{1}{2}(\omega^2) \frac{1}{2}$$

$$\text{Therefore } \omega = 36 \text{ rad/s}$$

135. **ANS – 1**

At the highest point let v is the speed

$$\frac{mv^2}{R} = mg + N$$

$$= N = \frac{mv^2}{R} - mg$$

R is min for first case so N_{max}